专题:近地表地下空间地球物理精细探测技术

连山关铀矿床地球物理特征及找矿预测研究

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摘要:连山关铀矿床是我国品位最富铀矿床之一。前人工作表明连山关岩体边部发育的韧性剪切带控制着铀矿床的产出,后期构造运动导致已富集的矿体被分割,使其连续性在空间上发生变化。为了进一步明确找矿靶区,本研究提出了多尺度地球物理探测技术,在综合研究区的地质、地球物理特征的基础上,针对连山关岩体接触带开展“预选区”到“有利区”再到“找矿靶区”的勘探技术路线。首先对航磁数据进行化极、向上延拓和方向导数等处理,总结出连山关铀矿床航磁 ΔT 化极后场强高于背景场 630～670nT,ΔT 上延 500m 后的场强高于背景场 450～505 nT,X、Y、Z 方向上的导数值区间分别为－0.2～0 nT/m,－0.03～0.03 nT/m,0.22～0.45 nT/m 的特征;对航放铀含量和土壤氡测量数据分别进行分区统计、确定异常下限,并进行了归一化处理,从而总结出连山关铀矿床的航放铀含量在 2.14×10^{-6}～2.34×10^{-6} g/g;归一化后相对值在 1.0～1.1;土壤氡浓度在 15×10^{3}～133.2×10^{3} Bq/m^{3},归一化后相对值在 1.5 ～13.3 的特征;综合上述磁场和放射场特征,建立了连山关铀矿床的地球物理-数学模型,结合研究区航磁、航放和土壤氡数据,运用半定量预测技术,在研究区内初步筛选出找矿有利区;最后,在深入分析、研究找矿研究区电性特征的基础上,利用连山关岩体接触带附近实测岩心资料,确定了岩体的解译标志,结合区域地质资料,对音频大地电磁测深数据进行了解译,最后在有利区内确定了找矿靶区 1 处,为连山关地区进一步铀矿地质工作提供了基础资料。

关键词:连山关铀矿床;地球物理特征;找矿预测

Study on geophysical characteristics of Lianshanguan uranium deposit and prospecting prognosis

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Abstract: Lianshanguan uranium deposit is one of the richest uranium deposit found in China.Previous works have proved that ductile shear zone developed at the edge of Lianshanguan rock mass controls uranium mineralization occurrence. In addition, late tectonic movement induces the separation of enriched ore body, which forms discontinuity in space. Based on the above situations, to further determine the target of prospecting, the multi-scale geophysical exploration technique is put forward, that is, based on the combining with geological data and geophysical characteristics in the study area, the technical route of prospecting from "candidate area" to "favorable area" and then to "prospecting target" is carried out in contact zone of Lianshanguan rock mass. That is to say, the aeromagnetic data is first processed with polarization, upward continuation and directional derivative, and it is concluded that the aeromagnetic ΔT post-polarization field strength of the Lianshanguan uranium deposit is higher than the background field by 630～670nT, and the field after ΔT is extended by 500m. The intensity is 450～505 nT higher than the background field, and the derivative value intervals in the X, Y, and Z directions are respectively －0.2～0nT/m, －0.03～0.03nT/m, 0.22～0.45nT/m; The released uranium content and the soil radon measurement data were separately statistically divided to determine the lower limit of anomalies, and normalized, so as to conclude that the airborne uranium content of the Lianshanguan uranium deposit was 2.14×10^{-6}～2.34×10^{-6} g/g; normalized relative value is 1.0～1.1; soil radon concentration is 15×10^{3}～133.2×10^{3} Bq/m3, normalized relative value is 1.5～13.3. And the radiometric characteristics at Lianshanguan deposit are summarized. And then, based on the above characteristics of magnetic field and radiometric field, the geophysics－mathematical model of Lianshanguan deposit is established. Combined with aeromagnetic, airborne radiometric and soil radon data, the favorable areas for ore prospecting are screened preliminarily by semi-quantitative prognosis technique in the study area. Then the characteristics of parameters of rock resistivity are analyzed and studied by using measured data of core near the contact zone of Lianshanguan rock mass. The interpretation guide of audio-magnetotelluric data is established. Combined with regional geological data, the interpretation of the measured audio-magnetotelluric data are carried out. In final, a prospecting target is predicted in contact zone of Lianshanguan rock mass, which provides the basic data for further uranium geological work in Lianshanguan area.

Keywords: Lianshanguan uranium deposit; Geophysical characteristics; Prospecting prediction